**MA3236 NONLINEAR PROGRAMMING**
Semester 1, AY 2026/2027
Tutorial 8

## Review: Lecture 16 — Lagrange–Newton and SQP

**From KKT to a Newton step.** For $\min f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) = \mathbf{0}$ with $C^2$ data, set $L = f + \boldsymbol{\lambda}^T\mathbf{g}$, $\mathbf{A} = [\nabla g_1, \ldots, \nabla g_m]$ and $\mathbf{H}_L = \nabla^2_{\mathbf{xx}} L$. With $\mathbf{z} = (\mathbf{x}, \boldsymbol{\lambda})$, linearize the KKT system $F(\mathbf{z}) = \mathbf{0}$ to obtain the Newton step

$$F(\mathbf{z}) = \begin{pmatrix} \nabla_{\mathbf{x}} L \\ \mathbf{g} \end{pmatrix}, \qquad \underbrace{\begin{pmatrix} \mathbf{H}_L & \mathbf{A} \\ \mathbf{A}^T & 0 \end{pmatrix}}_{K(\mathbf{z})} \begin{pmatrix} \mathbf{d} \\ \boldsymbol{\delta} \end{pmatrix} = -F(\mathbf{z}), \qquad \mathbf{z}^+ = \mathbf{z} + (\mathbf{d}, \boldsymbol{\delta}).$$

- LICQ (rank $\mathbf{A} = m$) and $\mathbf{d}^T\mathbf{H}_L\mathbf{d} > 0$ for all $\mathbf{0} \neq \mathbf{d} \in \mathrm{Null}(\mathbf{A}^T)$ make $K$ nonsingular; positive definiteness on all of $\mathbf{R}^n$ is unnecessary.
- At a KKT pair $\mathbf{z}^*$, nonsingular $K(\mathbf{z}^*)$ and locally Lipschitz $K$ give quadratic convergence from a sufficiently close start: $\|\mathbf{z}^{(k+1)} - \mathbf{z}^*\| \leq C\|\mathbf{z}^{(k)} - \mathbf{z}^*\|^2$.

**The same local step as an optimization problem.** Sequential quadratic programming (SQP) minimizes a quadratic model over linearized constraints:

$$\min_{\mathbf{d}} \ \nabla f^T\mathbf{d} + \tfrac{1}{2}\mathbf{d}^T\mathbf{B}\mathbf{d} \quad \text{s.t.} \quad \mathbf{g} + \mathbf{A}^T\mathbf{d} = \mathbf{0} \qquad (\text{multiplier } \widehat{\boldsymbol{\lambda}}).$$

- Under LICQ and the positive reduced curvature above, $\mathbf{B} = \mathbf{H}_L$ gives $\mathbf{d}_{\mathrm{SQP}} = \mathbf{d}_{\mathrm{LN}}$ and $\widehat{\boldsymbol{\lambda}} = \boldsymbol{\lambda} + \boldsymbol{\delta}$ by the QP's KKT equations. Thus $\widehat{\boldsymbol{\lambda}}$ is the new multiplier estimate, not an increment.
- For $\mathbf{h} \leq \mathbf{0}$, also impose $\mathbf{h} + \mathbf{C}^T\mathbf{d} \leq \mathbf{0}$, where $\mathbf{C} = [\nabla h_1, \ldots, \nabla h_p]$; the QP chooses its own active set and multipliers $\widehat{\boldsymbol{\nu}} \geq \mathbf{0}$.
- If the Hessian lacks positive reduced curvature, approximate or modify it; if the linearized constraints are inconsistent, use restoration or relaxation.

**Choose a step that makes actual progress.** The equation $\mathbf{g} + \mathbf{A}^T\mathbf{d} = \mathbf{0}$ need not give $\mathbf{g}(\mathbf{x} + \mathbf{d}) = \mathbf{0}$. For equality SQP, combine objective and constraint violation in

$$\phi_\rho(\mathbf{x}) = f(\mathbf{x}) + \rho\|\mathbf{g}(\mathbf{x})\|_1, \qquad \rho > 0.$$

If $\mathbf{B} \succ 0$, $\mathbf{d} \neq \mathbf{0}$ and $\rho > \|\widehat{\boldsymbol{\lambda}}\|_\infty$, then $D\phi_\rho(\mathbf{x}; \mathbf{d}) < 0$. Backtracking selects $\alpha > 0$ satisfying

$$\phi_\rho(\mathbf{x} + \alpha\mathbf{d}) \leq \phi_\rho(\mathbf{x}) + \sigma\alpha D\phi_\rho(\mathbf{x}; \mathbf{d}), \qquad 0 < \sigma < 1.$$

Trust regions and filters are alternative globalization strategies. Merit decrease measures progress, but does not certify KKT conditions.

**Check the original optimality conditions before stopping.** For $\mathbf{g} = \mathbf{0}$, $\mathbf{h} \leq \mathbf{0}$, use the infinity-norm residuals

$$r_{\mathrm{feas}} = \max\{\|\mathbf{g}\|_\infty, \|[\mathbf{h}]_+\|_\infty\}, \qquad r_{\mathrm{stat}} = \|\nabla f + \mathbf{A}\boldsymbol{\lambda} + \mathbf{C}\boldsymbol{\nu}\|_\infty,$$
$$r_{\mathrm{dual}} = \|[-\boldsymbol{\nu}]_+\|_\infty, \qquad r_{\mathrm{comp}} = \|\boldsymbol{\nu} \odot \mathbf{h}\|_\infty.$$

Require every applicable residual to be at most $\varepsilon_{\mathrm{out}}$; a small step alone is insufficient.

## Review: Lecture 17 — Barrier and primal interior point

**Keep inequality iterates strictly inside.** For $\mathbf{h}(\mathbf{x}) \leq \mathbf{0}$ and $\mu > 0$, retain equalities explicitly and solve

$$\min_{\mathbf{x}} \ \Phi_\mu(\mathbf{x}) = f(\mathbf{x}) - \mu \sum_{j=1}^{p} \log[-h_j(\mathbf{x})] \quad \text{s.t.} \quad \mathbf{g}(\mathbf{x}) = \mathbf{0}, \qquad \mathbf{h}(\mathbf{x}) < \mathbf{0}.$$

When all $h_j$ are convex, convex (strictly convex) $f$ makes $\Phi_\mu$ convex (strictly convex) on its domain.

**Read barrier stationarity as perturbed KKT.** At an interior point, define $\nu_j = \mu/[-h_j] > 0$. Then

$$\begin{aligned} \nabla \Phi_\mu &= \nabla f + \mathbf{C}\boldsymbol{\nu}, \\ \nabla^2 \Phi_\mu &= \nabla^2 f + \sum_j \nu_j \nabla^2 h_j + \sum_j \frac{\mu}{h_j^2} \nabla h_j \nabla h_j^T. \end{aligned}$$

Thus an exact barrier stationary pair $(\mathbf{x}, \boldsymbol{\lambda})$ satisfies

$$\nabla f + \mathbf{A}\boldsymbol{\lambda} + \mathbf{C}\boldsymbol{\nu} = \mathbf{0}, \quad \mathbf{g} = \mathbf{0}, \quad \mathbf{h} < \mathbf{0}, \quad \boldsymbol{\nu} > \mathbf{0}, \quad -\nu_j h_j = \mu.$$

- Along this central path, $r_{\text{comp}} = \mu$ and $-\boldsymbol{\nu}^T \mathbf{h} = p\mu$. With $C^1$ data, $\mu^{(k)} \downarrow 0$, $\mathbf{x}^{(k)} \to \mathbf{x}^*$ and LICQ at $\mathbf{x}^*$, the limit satisfies KKT. For nonconvex problems, minimality needs a separate check.
- If an active multiplier tends to $\nu_j^* > 0$, then $-h_j = \mu/\nu_j = \Theta(\mu)$ and $\mu/h_j^2 = \Theta(1/\mu)$. The Hessian formula therefore explains the large normal curvature near active constraints.

**Solve each barrier subproblem without leaving its domain.** With equalities, use the L16 Newton system with $f \leftarrow \Phi_\mu$; without them, solve $\nabla^2 \Phi_\mu \, \mathbf{d} = -\nabla \Phi_\mu$.

- For affine $h_j = \mathbf{a}_j^T \mathbf{x} - b_j$, restrict the trial step to

  $$\alpha_{\max} = \min \left\{ 1, \ \tau \min_{\mathbf{a}_j^T \mathbf{d} > 0} \frac{-h_j(\mathbf{x})}{\mathbf{a}_j^T \mathbf{d}} \right\}, \qquad 0 < \tau < 1,$$

  which preserves $\mathbf{h} < \mathbf{0}$ (empty minimum: $+\infty$). For nonlinear $\mathbf{h}$, check actual $\mathbf{h}(\mathbf{x} + \alpha \mathbf{d}) < \mathbf{0}$ before logs.
- For a descent direction without equalities, backtrack from the feasible trial step until

  $$\Phi_\mu(\mathbf{x} + \alpha \mathbf{d}) \leq \Phi_\mu(\mathbf{x}) + \sigma \alpha \nabla \Phi_\mu(\mathbf{x})^T \mathbf{d}, \qquad 0 < \sigma < 1.$$

  With equalities, use an equality merit function in the line search.

**Connect inner solves to the original problem.**

- At outer iteration $k$, warm start the inner solve to $\varepsilon_{\text{in}}^{(k)}$, infer $\nu_j = \mu/[-h_j]$, and test the original KKT residuals against $\varepsilon_{\text{out}}$. Otherwise reduce $\mu$ and coordinate $\varepsilon_{\text{in}}^{(k)}$; a small inner gradient or small $\mu$ alone is insufficient.
- To obtain a strict start, use Phase I: $\min_{\mathbf{x},t}\{t : \mathbf{g}(\mathbf{x}) = \mathbf{0}, \ h_j(\mathbf{x}) \leq t\}$. A feasible $t < 0$ supplies one; an empty strict interior precludes it.

1. **One Lagrange–Newton step and its SQP interpretation.** Consider

$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = (x_1-1)^2 + (x_2-1)^2 \quad \text{s.t.} \quad g(\mathbf{x}) = x_1 + x_2 + x_1^2 = 0.$$

   Start at $\mathbf{x}^{(0)} = (1,-2)^T$ and $\lambda^{(0)} = 0$.

   (a) Form and solve the Lagrange–Newton system. Find the full-step primal and multiplier updates.

   (b) Write and solve the equality SQP subproblem with $\mathbf{B}^{(0)} = \mathbf{H}_L^{(0)}$. Verify its multiplier's relation to the Newton multiplier update, and check both linearized and actual feasibility of the new point.

2. **A barrier path approaching an active constraint.** Let $\mathbf{a} = (1,2)^T$ and consider

$$\min_{\mathbf{x}\in\mathbf{R}^2} f(\mathbf{x}) = \|\mathbf{x}-\mathbf{a}\|_2^2 \quad \text{s.t.} \quad h(\mathbf{x}) = \mathbf{a}^T\mathbf{x} - 1 \le 0.$$

   (a) Form the logarithmic-barrier objective and state its domain. Find its unique minimizer $\mathbf{x}(\mu)$ for every $\mu > 0$.

   (b) Compute the inferred multiplier $\nu(\mu)$ and verify $-\nu(\mu)h(\mathbf{x}(\mu)) = \mu$. Find the limiting primal point and multiplier as $\mu \downarrow 0$, and check their KKT conditions.

3. **A feasible interior Newton step.** For the problem in Question 2, use $\mu = 1/10$ and $\mathbf{x}^{(0)} = \mathbf{0}$.

   (a) Evaluate $\nabla\Phi_\mu(\mathbf{x}^{(0)})$ and $\nabla^2\Phi_\mu(\mathbf{x}^{(0)})$, and compute the Newton direction $\mathbf{d}^{(0)}$. Does the full step stay in the barrier domain?

   (b) With $\tau = 9/10$, find the fraction-to-boundary trial length and point. At this point, infer $\nu$ and calculate $r_{\text{comp}}$. Is this point accurate enough for the outer method to stop with $\varepsilon_{\text{out}} = 10^{-3}$?

4. **Merit decrease after a curved-constraint step (challenge).** Return to Question 1 and its direction $\mathbf{d}^{(0)}$. Use $\phi_\rho = f + \rho|g|$, with a fixed $\rho > 0$.

   (a) Find $\phi_\rho(\mathbf{x}^{(0)} + \alpha\mathbf{d}^{(0)}) - \phi_\rho(\mathbf{x}^{(0)})$ for $0 < \alpha \le 1$. Which values of $\rho$ make the full step strictly decrease the merit function?

   (b) Take $\rho = 20$ and a backtracking rule that tries $\alpha = 1, 1/2, 1/4, \ldots$ until the merit value strictly decreases. Find the first accepted step. Explain, using the orders of the objective change and constraint violation, why sufficiently small positive steps work for every fixed finite $\rho$, even when the full step fails.